

Tree Recognition and Graph Edge Minimization from Anchored Distance Comparisons

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Abstract

We study whether comparisons of the form $d(o, x) < d(o, y)$ can be represented by shortest-path distances in a positively weighted tree whose vertices are exactly the observed objects. Two quadratic-time methods find candidate tree edges: Half-Space Proximal (HSP) selection and a minimum spanning tree of a metric satisfying the comparisons. If a realizing tree exists, both methods recover its edges. A separate linear program decides whether that tree admits suitable edge weights. Using distances from a root as variables gives at most four nonzero coefficients per inequality, bounded integer realizing weights, and small certificates of infeasibility for the candidate tree. If a tree realization exists, every additive triple possible in any compatible metric has its middle vertex on the corresponding tree path; the tree metric satisfies all these equalities at once. For general graphs, edges that can be omitted separately need not be omissible together. We give a smallest five-vertex example, a linear gap between the number of forced edges and the minimum number of edges, and the sharp worst-case minimum edge count $\binom{n}{2} - 1$. A seven-vertex example shows that maximizing the number of additive triples can require more edges than minimizing the edge count. Exact computational proofs and complete tree-recognition timings accompany the results.

1 Introduction and relation to previous work

An *anchored comparison* $d(o, x) < d(o, y)$ says that x is closer to the observed object o than y is. Such information may be available without numerical distances. We ask whether the comparisons can all be satisfied by shortest-path distances in a graph with positive edge weights. Every graph vertex must be an observed object. In particular, tree branch vertices cannot be hidden. For example, junctions in a communication graph might rank other junctions by path delay without reporting the delays themselves. This example motivates the model; it is not an empirical study.

For trees, there are two tasks: find the edges, then find weights that satisfy all comparisons. Complete anchored rankings determine at most one possible tree, but its weights may not exist. When they do exist, they also explain which triples can satisfy $d(u, w) = d(u, v) + d(v, w)$ in any metric with the same rankings. Such an equality is possible exactly when v is an internal vertex of the tree path from u to w , and a realizing tree metric satisfies all these equalities simultaneously. For general graphs, this simultaneous feasibility can fail. An equality through an intermediate vertex permits an edge to be omitted, but two individually possible omissions may require inconsistent distances. The paper gives the following results.

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1. *Testing tree edge weights.* Theorem 4.2 gives a linear program with $n - 1$ variables and at most four nonzero coefficients per inequality. Proposition 4.3 bounds integer realizing weights, and Corollary 4.4 bounds an infeasibility certificate for the candidate tree. HSP and a metric–MST (minimum spanning tree) method both find the candidate edges in quadratic time (Theorems 3.2 and 2.2).
2. *All possible additive triples for tree rankings.* Theorem 5.4 characterizes these triples by internal vertices of tree paths. Maximizing their number therefore recovers the same tree edges. The maximum count is easy to compute once the tree is known, although different edge weights can give the same rankings.
3. *Edges that cannot all be omitted.* Theorems 7.2 and 7.5 give a five-vertex example and a growing gap between the number of forced edges and the minimum number of edges. Proposition 7.6 proves that five vertices are necessary. Theorems 7.7 and 7.8 give the sharp worst-case edge count and show that maximizing additive triples and minimizing edges can have different solutions.

Previous work. Recovery and uniqueness of the tree edges already follow in our fully observed positive case from Guénoche’s order-distance theorem [8, Theorem 1]: every edge midpoint separates observed endpoints, so no edge contracts. His Theorem 2 and Section 4 also treat ordinal representability and numerical reconstruction; related split methods include [4, 14]. Moulton and Spillner study how the order-distance transformation behaves on sums of split distances, extending the tree setting. Our feasibility stage instead asks for one weighting preserving the original anchored rankings. Although the earlier order-distance calculation takes $O(n^4)$, this is not the best elementary baseline. Proposition 2.1 followed by dense Prim takes $O(n^2)$ and recovers the promised tree by its uniquely lightest cut edges. The original rankings still require a common positive weighting.

Prior method	Established results or methods	Role in this paper
HSP [5] and ordinal RNG [22, 12] Klauer [13]	Neighbor selection from anchored comparisons Graphs preserving distance orders, impossible additive triples, forced edges, maximum additive-triple count, tree uniqueness	Tree edge recovery, early rejection, comparison-query bounds Additive triples from anchored comparisons; different optima for the two objectives
MONA [15]	Exact distance orders and minimum edge count on observed vertices	Incompatible edge deletions and bounds on edge count
MAPNET / NETSCAL [16, 17]	Distinct removable endpoint pairs; greedy initialization and weight fitting	Exact feasibility tests and proofs of optimality

These precedents use different observation and fitting conventions. Klauer’s ordinal distance variables are synthesized, not supplied measurements. His completeness result for forbidden triples assumes a stronger ordering of object pairs; our proof using the common vertex of three tree paths uses anchored comparisons alone. MONA explicitly permits zero ordinal error and minimizes edges before an optional weight objective [15, pp. 299–302]. It describes exact solution through betweenness filtering, compatible path alternatives, and iterative integer weights, without a detailed termination or complexity theorem in that chapter. Its global pair ranks contain more information than our anchored input, but the edge-minimization objective is already the same.

Klauer–Carroll’s fixed-metric theorem counts *distinct endpoint pairs* with additive witnesses [16, p. 250]. Their interval-scale fitting algorithm combines greedy deletion starts with continuous

penalty optimization and can reach local optima [16, pp. 252–255]. NETSCAL’s symmetric ordinal criterion is exactly the relative neighborhood graph condition [17, Theorem 2]; its subsequent numerical fit uses a power transformation of measured proximities. Here we allow any positive weights satisfying the anchored comparisons and require exactly checkable feasibility and optimality proofs. We do not claim that combining greedy edge selection with weight fitting is new.

Model boundaries. Partial interleaf orders and ordinal phylogenies are studied in [9, 10, 11]; permitting latent tree vertices leads to a different, NP-complete problem [1]. Our partial-data theorem requires explicit coverage of every nonedge (Theorem 6.2); it does not solve arbitrary partial reconstruction. Numerical cactus realization [18] permits auxiliary vertices and minimizes total weight. Circular seriation [19] supplies ordering structure but is distinct from finding a common weighting. Finally, nearest-neighbor navigation [7, 6, 25, 26] preserves greedy progress under an input distance function, whereas we require exact shortest-path rankings. Neither navigation results nor general maximum-feasible-subsystem hardness [24] settle the complexity of our graph objective.

2 The model and an elementary metric consistency test

Let V have $n \geq 2$ elements. Our input is a set of *anchored comparisons* $x \prec_o y$, meaning that x is closer to o than y is. In the complete strict case, these comparisons specify a total order at each anchor. We store them as *anchored rankings* $\Pi = (\pi_o)_{o \in V}$, where π_o is a permutation of $V \setminus \{o\}$. We call one such collection a *ranking input*. This is the object often called a *profile* in earlier terminology. Write $x \prec_o y$ when x precedes y in π_o . A metric d *realizes*, or is *compatible with*, Π if

$$x \prec_o y \iff d(o, x) < d(o, y).$$

A *tree realization* is a tree $T = (V, E)$ with $w \in \mathbb{R}_{>0}^E$ whose path metric realizes Π . We also call such a metric a *tree metric*. Throughout, a *positive tree* means a tree with positive edge weights. Its vertices are exactly V unless stated otherwise. Recovering the tree means finding its edge set E ; finding the edge weights is a separate task. Since the vertex set is fixed, we identify graphs with their edge sets in set equalities and inclusions. Earlier papers may refer to the graph as a *network*, its edge structure as its *topology*, and the observation of every vertex as a *fully labeled* model. The one-vertex case has the unique empty realization and is handled separately. For $n = 2$, the single edge with any positive length realizes the unique ranking input.

We first test whether any metric can satisfy all anchored comparisons. Passing this test gives a compatible metric, but leaves open whether a positive tree can satisfy the same comparisons. Form a directed graph L_Π whose vertices are unordered pairs $\binom{V}{2}$. For each consecutive pair x, y in π_o , add the arc $\{o, x\} \rightarrow \{o, y\}$.

Proposition 2.1 (Metric consistency) *A complete strict ranking input is metric-realizable if and only if L_Π is acyclic. This can be decided, and a realizing integer metric constructed, in $O(n^2)$ time and storage. For a partial set Q of strict comparisons, the same assertion holds in $O(n^2 + |Q|)$ time using one arc per observation.*

Proof. A directed cycle requires a distance to be strictly smaller than itself. Conversely, let $e_1, \dots, e_m$, $m = \binom{n}{2}$, be a topological ordering. Set $d(e_j) = m + 1 + j$ and $d(u, u) = 0$. All required inequalities hold. Every off-diagonal distance is at least $m + 2$ and at most $2m + 1$, so the sum of any two is larger than every individual distance. Thus all triangle inequalities hold. The graph has $O(n^2)$ vertices and arcs in the complete case; ordinary topological sorting gives the stated bounds. $\square$

The equivalence between ordinal symmetric-distance compatibility and metric compatibility is already implicit in Klauer’s existence characterization [13, Theorem 1] and is recorded, for example, in Appendix A of Victor et al. [23]. Proposition 2.1 is its elementary finite algorithmic form, not a separate claim of geometric novelty. It also shows why triangle-inequality LPs are unnecessary merely to find an initial compatible metric. Cycles within individual rank rows are not the only obstruction: the pair-order graph checks consistency across anchors.

Theorem 2.2 (Compatible-metric MST baseline) *If a complete strict ranking input Π is realized by a positive tree T on exactly V , then T is the unique minimum spanning tree of every symmetric dissimilarity δ realizing Π . Consequently a compatible metric from Proposition 2.1, followed by dense Prim, gives an $O(n^2)$ algorithm for finding candidate tree edges. Testing its candidate with the fixed-tree feasibility system is an exact recognition algorithm.*

Proof. Fix $uv \in E(T)$ and any other pair ab crossing the cut of $T - uv$, oriented so that its path runs $a \cdots u - v \cdots b$. Tree positivity and the anchored comparisons imply

$$\delta(u, v) \leq \delta(u, b) \leq \delta(a, b).$$

The first inequality is strict if $b \neq v$ (anchor u), and the second is strict if $a \neq u$ (anchor b); otherwise the corresponding pairs are identical. Since $ab \neq uv$, at least one is strict. Hence uv is uniquely lightest across its cut. The MST cut property forces every edge of T , proving uniqueness. The construction and dense Prim each take $O(n^2)$ time. If the subsequent weight test succeeds it supplies a realization; if any realizing tree existed, the cut argument would have selected it. A failed weight test therefore rejects correctly. $\square$

The constructed metric need not be a tree metric; it is used only to choose the candidate tree edges. No measured distances or comparisons of disjoint pairs are required; the pair-order completion is constructed from the input rankings. Unlike HSP with a selection limit, Prim always returns a tree on metric-compatible input, so rejection of such an input occurs at the weight stage.

3 Recovery of tree edges by HSP with a selection limit

The metric–MST baseline already selects the only possible realizing tree under the promise. We now obtain the same candidate directly by ordinal HSP screening, which also provides early rejection and an oracle-query bound. Neither candidate procedure certifies the existence of its weights.

At anchor o , initialize an active set $S = V \setminus \{o\}$. Repeatedly select its first element a in π_o , output $o \rightarrow a$, and remove a and each remaining x satisfying $a \prec_x o$. This is the ordinal form of HSP. Write $\text{HSP}^\rightarrow(\Pi)$ for its directed output and $P(\Pi)$ for its undirected symmetrization. The *relative neighborhood graph* (RNG) is defined by

$$uv \in \text{RNG}(d) \iff \nexists z \notin \{u, v\} : d(u, z) < d(u, v), \ d(v, z) < d(u, v).$$

A vertex z satisfying both strict inequalities is a witness for excluding uv from the RNG (an *open-lune witness* in the literature). Thus the RNG depends only on the anchored comparisons. We use this strict version of the definition throughout.

Theorem 3.1 (HSP and RNG recover the tree edges) *If d is the metric of a positive tree $T = (V, E)$ and its anchored rankings are strict, then*

$$\text{HSP}^\rightarrow(\Pi_d) = \{(u, v), (v, u) : uv \in E\}, \quad P(\Pi_d) = \text{RNG}(d) = T.$$

The tree is also the unique minimum spanning tree of the complete graph weighted by d .

Proof. Fix o . Removing o partitions the tree into branches, one for each neighbor v . Positivity makes v the unique nearest point to o in that branch. Selecting a point a from another branch cannot screen any point x in this branch, since $d(x, a) = d(x, o) + d(o, a) > d(x, o)$. Therefore the first selection in this branch is v . It screens every deeper vertex x because $d(x, v) = d(x, o) - w_{ov} < d(x, o)$. Exactly the neighbors of o are selected.

A nonedge uv has an internal path vertex closer to both endpoints than they are to each other. For a tree edge, every third vertex lies on one side of its fundamental cut and is farther than $d(u, v)$ from the other endpoint. This proves the RNG identity. Finally, uv is the unique shortest pair across the cut of $T - uv$: every other crossing path contains uv and at least one additional positive edge. The cut property forces every minimum spanning tree to contain every edge of T . $\square$

For example, on the weighted path $b \overset{2}{-} a \overset{1}{-} o \overset{4}{-} c \overset{8}{-} d$, the row at o is (a, b, c, d) . Selecting a removes b ; selecting c removes d . The outgoing arcs are exactly $o \rightarrow a$ and $o \rightarrow c$.

3.1 Explicit rankings

Store inverse ranks $\rho_o(x)$. A screening test is the constant-time lookup $\rho_x(a) < \rho_x(o)$. Algorithm 1 imposes a global selection bound, so the running time remains quadratic even on rejected inputs.

Algorithm 1. HSP edge recovery with a selection limit

1. Validate each rank row and build all inverse ranks. Initialize $D = \emptyset$.
 2. For each $o \in V$, mark all $x \neq o$ active and scan π_o once in order.
 3. When the scanned entry a is active, insert (o, a) in D . If $|D| = 2n - 1$, return REJECT immediately.
 4. Mark a inactive. Scan the other vertices x and mark x inactive if it is active and $\rho_x(a) < \rho_x(o)$. Continue the rank-row scan.
 5. After all anchors, check reciprocity of D and that its symmetrization is connected with $n - 1$ edges. Return this tree, or REJECT.
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Theorem 3.2 (Quadratic recovery of candidate tree edges) *Algorithm 1 takes $O(n^2)$ time and storage on every valid input. It never rejects a ranking input from a positive tree on the observed vertices. If it returns a tree, every realizing tree must have those edges.*

Proof. All rank-row scans and active-set initializations cost $O(n^2)$. Before the selection limit is reached, at most $2n - 2$ screening passes have occurred, each costing $O(n)$. Reciprocity and connectivity are checked within the same bound. Theorem 3.1 proves validity of the selection limit and uniqueness. $\square$

Passing this test is necessary, but need not establish realizability even when the ranking input is known to come from a metric; see Example 4.5.

3.2 Comparison-oracle access

Suppose a binary oracle answers $\mathcal{O}(o; x, y) = \mathbf{1}[x \prec_o y]$. An active minimum can be found by a tournament using at most $n - 2$ queries; screening requires at most another $n - 2$.

Corollary 3.3 (Comparison queries for recovering a promised tree) *Under a promise of a positive tree on the observed vertices, the edge set can be recovered with at most $4(n - 1)(n - 2) < 4n^2$ binary queries and $O(n)$ auxiliary storage.*

Proof. There are exactly $2n - 2$ selections. Multiply by the bound $2(n - 2)$ per selection. The active set and tree output require $O(n)$ storage. $\square$

This improves the constant in the selection-limit estimate by counting only selections that actually occur under the promise. It does not give complete unpromised recognition with the same query bound. Sorting every row and then using the complete algorithm gives an $O(n^2 \log n)$ query upper bound for unpromised recognition. Conversely, the n^{n-2} labeled trees each admit strict positive weights (assign distinct powers of two to their edges), so any exact binary decision tree for recovering the edges has worst-case depth at least $(n - 2) \log_2 n$.

4 Testing edge weights by linear programming

Recovery of tree edges reduces the tree question to one common system of weight inequalities. This section gives that system and explicit certificates for either outcome. Here a *certificate* is explicit data that can be checked using exact arithmetic: suitable weights prove feasibility, while a nonnegative combination of inequalities proves infeasibility. Example 4.5 shows why this second stage is necessary even for a metric-compatible ranking input.

Fix the candidate tree and root it at r . Let $p(v)$ be the parent of $v \neq r$ and put $h_r = 0$. Positive edge weights correspond bijectively to root-distance variables $h_v = d_T(r, v)$ with $h_v - h_{p(v)} > 0$, via

$$w_{p(v)v} = h_v - h_{p(v)}, \quad D_h(u, v) = h_u + h_v - 2h_{\text{lca}(u, v)}.$$

For $\pi_o = (v_1^o, \dots, v_{n-1}^o)$, it is enough to require the $n - 2$ consecutive gaps to be positive. For one comparison $x \prec_o y$, the row is

$$g_{o;x,y}(h) = h_y - h_x - 2h_{\text{lca}(o,y)} + 2h_{\text{lca}(o,x)} > 0. \quad (1)$$

Let $A = A_T(\Pi)$ contain all $n - 1$ positivity rows and all $n(n - 2)$ consecutive rows, omitting the root column.

Lemma 4.1 (Coefficient bounds) *After collecting coincident positions, every row of A has support at most four, entries in $\{0, \pm 1, \pm 2\}$, and squared Euclidean norm at most 10.*

Proof. Write $a = \text{lca}(o, x)$ and $b = \text{lca}(o, y)$. If $a = b$, their terms cancel, leaving $h_y - h_x$. Otherwise a, b are distinct ancestors of o and are comparable. A coincidence $y = a$ would give $b = \text{lca}(o, a) = a$, a contradiction; similarly $x = b$ is impossible. The only remaining coincidences involving the endpoints are $x = a$ and $y = b$. They replace, respectively, $-1 + 2$ by 1, or $1 - 2$ by -1 . Thus no coefficient of magnitude three occurs, and the largest possible squared norm is $1 + 1 + 4 + 4 = 10$. Removing the root column can only decrease the norm. Positivity rows satisfy the same bounds. $\square$

Theorem 4.2 (Testing the candidate tree weights) *A complete strict ranking input has a positive tree realization on the observed vertices if and only if Algorithm 1 returns a tree T and $A_T(\Pi)h > 0$ is feasible. The matrix has $p = n - 1$ columns, $q = n^2 - n - 1$ rows and $O(n^2)$ nonzeros. It can be constructed in $O(n^2)$ time. Recognition and construction are strongly polynomial in the arithmetic model.*

Proof. Positivity rows give positive weights, and (1) is exactly the corresponding distance difference. All consecutive gaps being positive is equivalent to the entire rank row being realized. Conversely

a realizing weighting provides these root-distance variables, and its edge set is recovered by Theorem 3.1.

Constant-time LCA queries after linear preprocessing [2] construct every row in constant time. Alternatively, an Euler-tour sparse table uses $O(n \log n)$ preprocessing and constant query time, still within the overall quadratic bound.

Homogeneity gives $Ah > 0$ if and only if $Ah \geq \mathbf{1}$. Split free variables as $h = h^+ - h^-$ and introduce slacks $s \geq 0$ to obtain

$$Ah^+ - Ah^- - s = \mathbf{1}, \quad h^+, h^-, s \geq 0.$$

The standard-form constraint matrix still has entries bounded in magnitude by two. Tardos's bounded-matrix linear-programming result [21] therefore applies. The number of arithmetic operations is polynomial in n ; it is not claimed to be quadratic. $\square$

Later developments connect the classical complexity theorem to modern approximate solvers. Dadush, Natura, and Véggh [27] obtain exact primal and dual solutions by combining approximate LP solves with constructive proximity arguments, with complexity controlled by matrix condition measures. For Theorem 4.2, Tardos's original bounded-matrix result already gives the asserted strongly polynomial bound. Obtaining a sharper runtime from the later framework would require a separate analysis of our standard-form matrices and subroutines; rows with at most four nonzero coefficients alone do not supply such a bound.

Strongly polynomial feasibility does not depend on using root-distance variables. In ordinary edge coordinates each distance difference is a difference of two path-incidence vectors, so coefficients lie in $\{0, \pm 1\}$; bounded-matrix LP already applies there. The root-distance formulation instead gives at most four nonzeros per row, the uniform squared row-norm bound of ten, quadratic assembly, and the explicit arithmetic bounds below. An edge-coordinate row can involve $\Theta(n)$ edges.

Proposition 4.3 (Bounded integer edge weights) *Every accepted ranking input has a realizing integer weighting with each weight at most $\lceil 11^{(n-1)/2} \rceil$. Thus each weight needs $O(n)$ bits.*

Proof. The feasible polyhedron $Ah \geq \mathbf{1}$ has no lines, since the positivity rows form an invertible tree-incidence system after fixing the root. More concretely, all nonroot root-distance variables are positive; minimizing their sum has a nonempty compact sublevel set and attains an optimum at a vertex. At a vertex, some nonsingular $p \times p$ row submatrix B satisfies $Bh = \mathbf{1}$. Let $D = |\det B| \geq 1$ and set $H = Dh$. Then H is integral and $AH \geq D\mathbf{1}$. By Cramer's rule, H_i is, up to sign, the determinant obtained by replacing one column of B by ones. Each row of this matrix has squared norm at most 11, so Hadamard's inequality gives $|H_i| \leq 11^{p/2}$. Positivity gives $0 < H_v - H_{p(v)} \leq H_v$, proving the weight bound. $\square$

A generic Hadamard estimate in edge coordinates, where a row may have norm $\Theta(\sqrt{n})$, yields only an $O(n \log n)$ bit bound for such integral realizations. The constant squared row norm here improves this estimate to $O(n)$ bits per weight. The same comparison applies to the minor bounds for multipliers below. Sharpness of the exponential weight and multiplier bounds, including their bases and the $O(n)$ bit order, is not established here; they are explicit sufficient bounds.

4.1 Infeasibility certificates

Gordan's alternative states that exactly one of $Ah > 0$ and

$$A^\top \lambda = 0, \quad \lambda \geq 0, \quad \lambda \neq 0 \tag{2}$$

is feasible; see [20]. Indeed, after normalizing $\mathbf{1}^\top \lambda = 1$, the second statement is that zero lies in the convex hull of the rows. If it does not, strict separation gives the first statement. The two cannot both hold because $\lambda^\top Ah$ would be both zero and positive.

Corollary 4.4 (A small infeasibility certificate) *If the weight phase fails, there is an integral certificate (2) supported on at most n rows. Each nonzero multiplier can be bounded by $\lceil 10^{(n-1)/2} \rceil$.*

Proof. Carathéodory's theorem gives a positive dependence on at most $p + 1 = n$ rows. Choose an inclusion-minimal positive support of size k . Its row rank is $k - 1$: a second independent dependence would allow perturbation of the positive vector until one coefficient vanishes. For $k > 1$, choose $k - 1$ independent coordinate columns and take the signed maximal minors of the resulting $k \times (k - 1)$ matrix. They give an integral dependence with a common sign, hence positive after a possible sign change. Each minor has magnitude at most $10^{(k-1)/2}$ by Lemma 4.1. A zero row, if present, has the one-row certificate with multiplier one. $\square$

The support bound is conditional on the candidate tree. It is not a claim that at most n raw comparisons alone exclude every possible edge set. A full rejection certificate includes the edge recovery, which can be replayed in $O(n^2)$ time, and, when needed, the displayed positive dependence. For extraction, solve $A^\top \lambda = 0$, $\mathbf{1}^\top \lambda = 1$, $\lambda \geq 0$ and take a basic feasible solution or reduce its support.

Example 4.5 (A candidate star with no suitable weights) The following integer matrix is a metric with strict anchored rankings:

$$d = \begin{pmatrix} 0 & 10 & 11 & 12 \\ 10 & 0 & 13 & 15 \\ 11 & 13 & 0 & 14 \\ 12 & 15 & 14 & 0 \end{pmatrix}.$$

All off-diagonal entries lie between 10 and 15, so the triangle inequalities are immediate. Its rankings are

$$(1, 2, 3), \quad (0, 2, 3), \quad (0, 1, 3), \quad (0, 2, 1).$$

HSP returns the star centered at 0: the center screens all other leaves at each leaf anchor, while no leaf screens another at anchor 0. Let the three star weights be a, b, c . Anchor 0 requires $b - a > 0$, whereas anchor 3 requires $a - b > 0$. These two consecutive rows sum to zero. Thus even symmetric metric consistency together with a valid HSP tree does not replace the weight test.

Example 4.6 (An accepted instance) Take edges 01, 12, 13, 34, 35 with weights 2, 3, 5, 11, 17, respectively. With root 0 the root-distance variables are $(h_1, h_2, h_3, h_4, h_5) = (2, 5, 7, 18, 24)$, and the ranking input is

o	π_o	o	π_o
0	(1, 2, 3, 4, 5)	3	(1, 0, 2, 4, 5)
1	(0, 2, 3, 4, 5)	4	(3, 1, 0, 2, 5)
2	(1, 0, 3, 4, 5)	5	(3, 1, 0, 2, 4)

For instance, at anchor 3, the consecutive comparison $0 \prec_3 2$ becomes $h_2 - 2h_1 > 0$, with value 1. At anchor 2, the comparison $0 \prec_2 3$ becomes $h_3 - 2h_1 > 0$, with value 3. Positivity and all other consecutive rows are checked in the same way.

5 Additive triples, tree edges, and computational cost

A successful weight test also restricts every other metric compatible with the rankings. We now ask which additive-triple relations those metrics can exhibit. The next argument uses the common vertex of three tree paths to show that trees allow all individually possible additive triples to coexist, linking recognition to the edge-deletion objective introduced below.

For a metric-realizable ranking input, let $s(\Pi)$ be the minimum number of edges in a positive weighted graph on vertex set V realizing Π . For a fixed metric d , call a pair uv *required* if no intermediate vertex z satisfies $d(u, v) = d(u, z) + d(z, v)$. Define the *distance-preserving graph* $B(d)$ to contain exactly these pairs:

$$uv \in E(B(d)) \iff d(u, v) < d(u, z) + d(z, v) \text{ for all } z \notin \{u, v\}.$$

Proposition 5.1 shows that $B(d)$, with edge lengths from d , preserves every distance and is contained in every graph on V that does so. It is therefore the unique smallest graph for this *fixed metric*. This graph is called the *basic graph* in the earlier reduction of Klauer [13, Sections 4 and 6] (see also [3]). The distinct-pair counting characterization is explicit in [16, p. 250], with a directed version and deletion proof in [17, Theorem 1 and Appendix]. We give its short proof to fix the model.

Proposition 5.1 (The minimum graph preserving a fixed metric) *Giving each edge uv of $B(d)$ length $d(u, v)$ realizes d . Every weighted graph on V realizing d contains $B(d)$. Consequently,*

$$s(\Pi) = \min_{d \text{ realizing } \Pi} |E(B(d))|, \quad s(\Pi) = n - 1 \iff \Pi \text{ has a tree realization.}$$

Proof. A removable pair has $d(u, v) = d(u, z) + d(z, v)$. Each summand is strictly smaller than $d(u, v)$, so recursive decomposition terminates in required pairs, even if unrelated distances tie. Their concatenation has total length $d(u, v)$; triangle inequalities rule out a shorter route. Conversely, omitting a required pair would force an internal vertex on a shortest path, giving just such a tight triangle, a contradiction. The last equivalence uses the fact that a connected n -vertex graph has at least $n - 1$ edges, with equality exactly for trees. $\square$

Theorem 5.2 (HSP edges occur in every realizing graph) *For every compatible metric d , $P(\Pi) \subseteq B(d)$. Hence every realizing graph on V contains the symmetrized HSP graph.*

Proof. Suppose a selected arc $o \rightarrow v$ has a tight witness z , so $d(o, v) = d(o, z) + d(z, v)$. Point z precedes v in the anchor- o order. Before v is selected, z was selected itself or screened by an earlier selected a with $d(z, a) < d(z, o)$; put $a = z$ in the former case. Then

$$d(v, a) \leq d(v, z) + d(z, a) < d(v, z) + d(z, o) = d(v, o).$$

Thus a would have screened v , a contradiction. $\square$

A *forced edge* is present in every graph on V whose shortest-path distances satisfy the rankings. The set of forced edges is

$$\mathcal{F}(\Pi) = \bigcap_{d \text{ realizing } \Pi} E(B(d)).$$

A required pair is defined for one fixed metric; a forced edge is required for every compatible metric. Even knowing all forced edges need not determine the minimum edge count.

The forcing proof is an ordinal forbidden-triple certificate in the sense of Klauer [13, Theorem 2]: a competing triangle and anchored inequalities force strict slack. HSP efficiently collects such certificates; the general forcing principle is established prior work.

5.1 Additive triples when a tree realization exists

A triple (u, v, w) of distinct vertices is *additive* for d if $d(u, w) = d(u, v) + d(v, w)$. Thus a shortest route from u to w can pass through v . A vertex v satisfying this equality is an *additive witness* for the endpoint pair uw . We identify (u, v, w) with (w, v, u) by requiring $u < w$ in an arbitrary fixed label order. Different middle vertices for the same endpoints are counted separately. For a compatible metric define

$$\mathcal{A}(d) = \{(u, v, w) : u < w, v \notin \{u, w\}, d_{uw} = d_{uv} + d_{vw}\}, \quad U(\Pi) = \bigcup_{d \models \Pi} \mathcal{A}(d).$$

This convention counts each additive triple once. Klauer's ordered-triple count doubles this variable part; including repeated-vertex triples adds a constant. Neither convention changes maximizers. Let $\text{Bet}(T)$ contain the triples whose middle vertex lies internally on their endpoint path. This is often called the tree's *betweenness relation*; here it simply records which vertices lie between two others on a tree path.

The *median* of three vertices in a tree is the unique vertex common to the three paths joining each pair. It is a vertex of the tree, not a median distance value. The next lemma uses the fact that this vertex is observed, so the comparisons anchored there or referring to it are available.

Lemma 5.3 (An off-path vertex cannot give an additive triple) *Suppose a positive tree T on exactly V realizes Π . If v is not on the u – w path, then every compatible metric d satisfies $d_{uw} < d_{uv} + d_{vw}$.*

Proof. Let m be the median of u, v, w in T . Since v is off the endpoint path, $m \neq v$, and positivity gives $d_T(u, m) < d_T(u, v)$ and $d_T(w, m) < d_T(w, v)$. If m is an endpoint, the corresponding comparison follows from a zero diagonal and positivity. Otherwise these are observed anchored comparisons. They hold in every compatible d , so

$$d_{uw} \leq d_{um} + d_{mw} < d_{uv} + d_{vw}.$$

The argument uses the observed median, not a numerical distance query. □

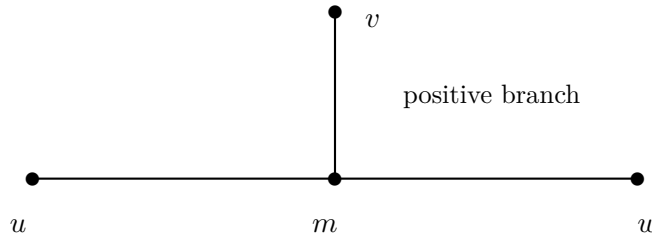


Figure 1: A vertex outside the endpoint path cannot give an additive triple. The anchor u ranks m before v , and so does w . In every compatible metric, the triangle inequality through the observed m then makes the route through v strictly longer than $d(u, w)$. The proof also covers $m = u$ or $m = w$.

Theorem 5.4 (Additive triples determine the realizing tree) *If a positive tree T on exactly V realizes Π , then*

$$\mathcal{A}(d) \subseteq \text{Bet}(T) \quad (d \models \Pi), \quad U(\Pi) = \text{Bet}(T).$$

Every maximizer of $|\mathcal{A}(d)|$ satisfies

$$\mathcal{A}(d) = \text{Bet}(T), \quad B(d) = T = P(\Pi) = \mathcal{F}(\Pi).$$

Its metric is realized on T with positive edge lengths d_{uv} .

Proof. Lemma 5.3 excludes all triples outside $\text{Bet}(T)$. The realizing tree metric attains every member of that finite set, so the union is $\text{Bet}(T)$ and every cardinality maximizer attains the whole set. A tree edge has no internal path vertex, whereas every nonedge does; hence $B(d) = T$. Repeated on-path equalities decompose each distance into the lengths of the tree edges on its path. Theorems 3.1 and 5.2, together with $B(d_T) = T$, give the HSP and forced-edge set identities. $\square$

This compares tree recovery from anchored rankings with reconstruction by maximizing the number of additive triples. It does not require comparisons between disjoint endpoint pairs. It does not assert uniqueness of weights: the observed three-vertex path with successive weights $(1, 2)$ and $(1, 3)$ gives the same rankings. Nor does it assert that every compatible metric reduces to the tree. Adding a fixed $c > 0$ to every off-diagonal tree distance preserves all rankings and makes every triangle on distinct vertices strict, so the distance-preserving graph becomes complete.

Corollary 5.5 (Counting additive triples from the tree edges) *Once T is recovered, the maximum number of additive triples is computable in $O(n)$ additional time as*

$$\max_{d \models \Pi} |\mathcal{A}(d)| = \sum_{e \in E(T)} s_e(n - s_e) - \binom{n}{2},$$

where s_e is the size of a component of $T - e$.

Proof. Each endpoint pair contributes its path length in edges minus one. Summing path-edge incidences over cuts gives the displayed expression; subtree sizes are computed in linear time. $\square$

5.2 A fair computational comparison

On the tree promise, HSP with a selection limit and compatible-metric MST both return the realizing tree edges in $O(n^2)$ time. Either can then compute the optimum number of additive triples in $O(n)$ additional time by Corollary 5.5. The MST baseline constructs a synthetic metric, but needs no measured distances. HSP offers the separate comparison-oracle guarantee, forced-edge information beyond trees, and rejection before solving for weights. An asymptotic quadratic advantage over this baseline is not claimed. Reporting all additive triples is a different output task: a path has $\binom{n}{3}$ such triples, whereas the tree has only $n - 1$ edges.

A direct distance formulation has $\binom{n}{2}$ numerical variables and $3\binom{n}{3}$ triangle inequalities, with at most $\binom{n}{3}$ candidate tight triples after the elementary ordinal orientation test. The system in root-distance variables instead has $n - 1$ variables and $n^2 - n - 1$ sparse rows. This is a formulation-size comparison, not a runtime theorem for Klauer’s original implementation and not an exponential lower bound on promised-tree optimization.

Indeed, an ordinal forbidden-triple implementation can also exploit the promise. For each candidate (u, v, w) , test whether some observed m satisfies $d_{um} \leq d_{uv}$ and $d_{wm} \leq d_{vw}$, with at least one strict comparison. Such an m excludes additivity. The median excludes every off-path

candidate, and no genuine path triple is excluded. Straight enumeration with inverse ranks takes $O(n^4)$ time and yields exactly $\text{Bet}(T)$. This bound describes that implementation, not a bound attributed to Klauer. A mixed integer search is unnecessary on the promised class. For unpromised recognition, a candidate tree still requires a weight-feasibility decision. Our LP supplies that decision; its strongly polynomial arithmetic complexity is not claimed to be quadratic.

6 Partial comparisons with enough evidence to exclude nonedges

The next result applies to incomplete observations when they provide enough evidence to exclude every nonedge of a candidate tree. It does not characterize all sets of partial comparisons that admit a tree.

Let Q be a set of strict observations $x \prec_o y$. Define

$$N_Q = \{uv : \exists z \notin \{u, v\}, (z \prec_u v) \in Q, (z \prec_v u) \in Q\}, \quad C_Q = (V, \binom{V}{2} \setminus N_Q).$$

For each pair uv in N_Q , an observed vertex is closer to both endpoints than they are to each other. Thus uv cannot be an RNG edge of a compatible metric.

Lemma 6.1 *The RNG of every finite metric is connected, including metrics with tied distances.*

Proof. Any minimum spanning tree is contained in the RNG. Otherwise an MST edge uv would have a witness z for exclusion from the RNG. Across the cut obtained by deleting uv , one of uz, vz crosses the cut and has strictly smaller weight, contradicting minimality. $\square$

Theorem 6.2 (Recognition when the candidate graph is a tree) *If C_Q is a tree, every metric realizing Q has RNG exactly C_Q . Moreover, Q has a positive tree realization on the observed vertices if and only if the positivity rows of C_Q and the rows (1) for all observations in Q are jointly feasible. Coverage and matrix construction take expected $O(n^2 + |Q|)$ time with hashed triple membership.*

Proof. For every compatible metric, $\text{RNG} \subseteq C_Q$. By Lemma 6.1, this subgraph is connected; a connected spanning subgraph of a tree is the entire tree. Any positive realizing tree has this RNG as its edge set. The sparse feasibility equivalence is then identical to the proof of Theorem 4.2; unobserved comparisons impose no constraints.

Hash Q . For each $(z \prec_u v) \in Q$, look up its mate $(z \prec_v u) \in Q$ and mark uv when present. Construct the complement and test that it is a tree in $O(n^2)$ additional time. Each observation contributes at most four matrix entries. $\square$

In particular, there is no need to separately query whether each candidate edge is protected against every third vertex. Connectedness of the RNG supplies this conclusion automatically once the complement is a tree.

Corollary 6.3 (A sufficient set of observed comparisons) *For every positive weighted tree $T = (V, E)$, one can select a set Q of its anchored comparisons such that $C_Q = T$ and $|Q| \leq (n-1)(n-2)$.*

Proof. For each of the $\binom{n}{2} - (n-1)$ nonedges uv , select an internal path vertex z and record $z \prec_u v$ and $z \prec_v u$. These observations exclude every nonedge and no tree edge. There are at most $2[\binom{n}{2} - (n-1)] = (n-1)(n-2)$ observations. $\square$

If C_Q is disconnected, no compatible metric exists. If it is connected with cycles, coverage alone is inconclusive. For example, $Q = \{b \prec_a c, a \prec_b c\}$ on three labels admits two different paths. Nevertheless the candidate graph gives a bounded search when few extra edges remain.

Corollary 6.4 (Few excess candidate edges) *If C_Q is connected with $n - 1 + k$ edges, partial tree recognition can be performed using at most $\binom{n-1+k}{k}$ fixed-tree feasibility tests, besides polynomial work per test.*

Proof. Every realizing tree is a spanning tree of C_Q . Enumerate the deletion sets of size k , retain the sets whose complements are trees, and solve the corresponding partial systems. $\square$

For fixed k this is polynomial; its dependence $n^{O(k)}$ is an XP bound, not a fixed-parameter tractability claim.

7 Graphs with few edges and incompatible edge deletions

We return to complete anchored rankings. For general graphs, one realization may omit an edge and another may omit a different edge, even though every realization must retain at least one of them. We first give a five-vertex example, then show that the difference between the number of forced edges and the minimum edge count can grow linearly. We also determine the largest possible minimum edge count and compare edge minimization with maximization of additive triples.

7.1 The simultaneous-deletion criterion

Let $\mathcal{M}_1(\Pi)$ consist of compatible metrics normalized by $\sum_{uv} d(u, v) = 1$. For $S \subseteq \binom{V}{2}$ put

$$\mathcal{D}_S = \mathcal{M}_1(\Pi) \cap \bigcap_{uv \in S} \bigcup_{z \notin \{u, v\}} \{d : d(u, v) = d(u, z) + d(z, v)\}. \quad (3)$$

Proposition 7.1 *The graph $K_V - S$ admits positive weights realizing Π if and only if $\mathcal{D}_S \neq \emptyset$.*

Proof. For $d \in \mathcal{D}_S$, all pairs in S are removable, so $B(d) \subseteq K_V - S$. Give retained edges their metric lengths and apply Proposition 5.1. Conversely, every omitted pair has an internal vertex on a shortest path in a realizing graph; its normalized metric is in the corresponding tight-triangle sets. $\square$

A set S is a *minimal incompatible deletion set* if $\mathcal{D}_S$ is empty but $\mathcal{D}_{S'}$ is nonempty for every proper subset S' . The unions in (3) allow alternative witnesses; the intersection requires all deletions to use the same metric.

7.2 A five-vertex example requiring six edges

Theorem 7.2 *On $V = \{0, 1, 2, 3, 4\}$ consider*

o	π_o
0	(1, 3, 4, 2)
1	(0, 4, 3, 2)
2	(4, 3, 1, 0)
3	(0, 2, 4, 1)
4	(2, 1, 0, 3)

(4)

Its HSP edge set and forced-edge set both equal $C = \{01, 03, 14, 23, 24\}$. The set $\{04, 34\}$ is a minimal incompatible deletion set, and $s(\Pi) = 6 > |\mathcal{F}(\Pi)| = 5$.

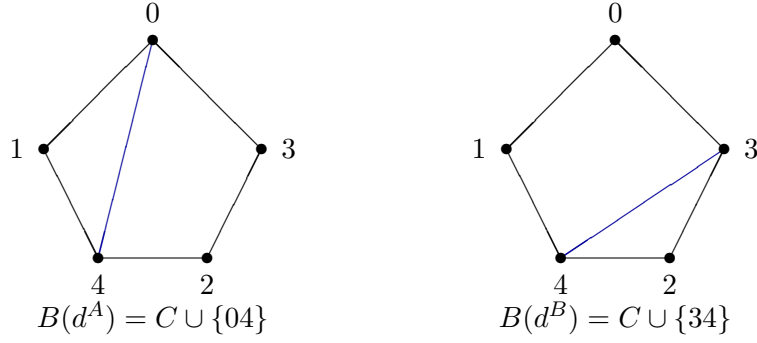


Figure 2: Two minimum realizations of the same anchored rankings. The five cycle edges are forced. Either highlighted diagonal can be omitted, but at least one must remain. Lengths are given in the proof.

Proof. Two integer metrics are specified by the following pair distances:

pair	01	02	03	04	12	13	14	23	24	34
d^A	6	15	7	11	14	13	10	8	4	12
d^B	3	16	9	10	13	12	7	10	6	11

They are the shortest-path metrics of the six-edge graphs $C \cup \{04\}$ and $C \cup \{34\}$, with retained edges carrying the displayed lengths. Direct shortest-path calculation gives the table, and sorting each row gives (4). Their distance-preserving graphs are exactly these respective six-edge graphs. HSP has outgoing neighbor sets

$$\{1, 3\}, \quad \{0, 4\}, \quad \{4, 3\}, \quad \{0, 2\}, \quad \{2, 1\}.$$

Theorem 5.2 forces C , while the intersection of the two displayed distance-preserving graphs is C . Thus the entire set of forced edges is C , and each of $04, 34$ can individually be deleted.

Suppose both could be deleted in one metric. Every tight-triangle witness must be closer to both endpoints than they are to each other. The only such witness for 04 is 1, so $d_{04} = d_{01} + d_{14}$. The possible witnesses for 34 are 0 and 2. Witness 0 is impossible: it would give

$$d_{34} = d_{03} + d_{04} > d_{03} + d_{01} \geq d_{13},$$

contrary to $d_{34} < d_{13}$. Therefore $d_{34} = d_{23} + d_{24}$. Write $a = d_{01}$, $b = d_{03}$, $c = d_{14}$, $t = d_{23}$, and $e = d_{24}$. The rankings and triangle inequalities imply

$$a + c = d_{04} < d_{34} < d_{13} \leq a + b, \quad t + e = d_{34} < d_{13} < d_{12} \leq c + e.$$

Hence $c < b$ and $t < c$, whereas anchor 3 requires $b < t$. This gives $c < b < t < c$, a contradiction. A five-edge realization would consist precisely of the forced set C and omit both pairs, so none exists. The displayed six-edge realizations attain the optimum. $\square$

7.3 Uniqueness can differ between the two objectives

For brevity, write uvw for an additive triple with middle vertex v ; the endpoint order in this notation is immaterial.

Proposition 7.3 *For the ranking input in Theorem 7.2, the maximum number of additive triples is six and its maximizing distance-preserving graph is uniquely $C \cup \{34\}$. There is also an inclusion-maximal feasible set of additive triples of size five.*

Proof. Of the ten ordinally eligible triples, the following are impossible:

$$\begin{aligned} d_{12} &\leq d_{14} + d_{24} < d_{13} + d_{23}, \\ d_{13} &\leq d_{01} + d_{03} < d_{14} + d_{34}, \\ d_{34} &< d_{13} \leq d_{01} + d_{03} < d_{04} + d_{03}. \end{aligned}$$

These exclude 132, 143, 304, respectively. The remaining set is

$$U = \{p, q, r, s, t, v, y\} = \{012, 032, 042, 014, 142, 103, 324\}.$$

The two metrics in Theorem 7.2 and the metric

$$(d_{01}, d_{02}, d_{03}, d_{04}, d_{12}, d_{13}, d_{14}, d_{23}, d_{24}, d_{34}) = (8, 42, 20, 26, 34, 28, 18, 22, 16, 27)$$

witness, respectively, the sets $\{q, r, t, v, y\}$, $\{p, r, s, t, v\}$, and $U \setminus \{y\}$. All distances, triangle inequalities, and rankings are checked directly. In particular every member of U is individually possible.

The incompatibility proof shows that s and y cannot coexist. Also p, r, t imply s , by substituting the three expressions for distances 02 and 12. A six-element subset must omit s or y ; omitting s would force it back through p, r, t , so the sole possible maximum is $U \setminus \{y\}$. Its endpoint pairs give $B = C \cup \{34\}$. The size-five set $\{q, r, t, v, y\}$ is maximal: adding s conflicts with y , and adding p forces s . It is not a cardinality maximum. $\square$

Proposition 7.4 (Two maximizing graphs) *The strict ranking input*

$$\begin{array}{c|c} 0 & (1, 2, 3, 4) \\ 2 & (0, 1, 4, 3) \\ 4 & (1, 0, 2, 3) \end{array} \quad \begin{array}{c|c} 1 & (0, 2, 4, 3) \\ 3 & (0, 1, 2, 4) \end{array}$$

has $P = \mathcal{F} = T_0 = \{01, 02, 03, 14\}$, but T_0 is not realizable. It has minimum edge count five and exactly two sets of additive triples of maximum size, each of size six, with distance-preserving graphs $T_0 \cup \{24\}$ and $T_0 \cup \{12\}$.

Proof. In lexicographic pair order, two compatible integer metrics are

$$d^E = (1, 3, 6, 7, 4, 7, 6, 9, 8, 13), \quad d^F = (4, 6, 10, 12, 7, 14, 8, 16, 15, 22).$$

Their distance-preserving graphs are the two stated graphs, and direct HSP gives T_0 . The forcing theorem and their intersection identify the forced-edge set. Writing the four tree-edge lengths as a, b, c, e for 01, 02, 03, 14, the rankings at 0 and 2 would require respectively $c < a + e$ and $b + a + e < b + c$. Thus no weighting of T_0 works, and the displayed five-edge realizations are minimum.

The individually possible triples are

$$U = \{a, b, c, d, e, f, g\} = \{014, 102, 103, 203, 214, 304, 314\}.$$

The other eligible triples 204, 213, 324 are excluded by

$$\begin{aligned} d_{24} &< d_{23} \leq d_{02} + d_{03} < d_{02} + d_{04}, \\ d_{23} &\leq d_{02} + d_{03} < d_{12} + d_{13}, \\ d_{34} &\leq d_{03} + d_{04} < d_{23} + d_{24}. \end{aligned}$$

The displayed metrics attain $U \setminus \{e\}$ and $U \setminus \{b\}$. Equalities a, b, e together imply the forbidden triple 204, so seven are impossible. A size-six subset omits one of a, b, e ; it cannot omit a , because c, f, g imply a by subtracting $d_{13} = d_{01} + d_{03}$ from $d_{34} = d_{03} + d_{04} = d_{13} + d_{14}$. The two remaining possibilities are exactly those witnessed. $\square$

7.4 A linear gap beyond all individually forced edges

Theorem 7.5 *For every integer $k \geq 1$, there is a metric-compatible strict ranking input on $n = 4k + 1$ vertices with*

$$|\mathcal{F}(\Pi)| = 5k, \quad s(\Pi) = 6k.$$

Thus $s(\Pi) - |\mathcal{F}(\Pi)| = (n - 1)/4$ can grow linearly.

Proof. Take k copies of Theorem 7.2, identify their vertices labeled 0 to one vertex r , and scale copy i by 100^i , for $i = 0, \dots, k - 1$. In each copy independently choose d^A or d^B . Within a copy retain its chosen metric; between distinct copies set

$$d(u, v) = d(u, r) + d(r, v).$$

This is the metric of the one-vertex union of the corresponding six-edge graphs.

All 2^k choices induce one common strict ranking input. To see this explicitly, the nonzero local distances are integers in $[3, 16]$, every anchored local gap has magnitude at least one, and the rankings from r are the same in both local metrics. Distances from a lower-scale copy to r are at most $16 \cdot 100^{i-1} < 100^i$, while the smallest nonzero distance in the next copy is $3 \cdot 100^i$. At an anchor within copy i , comparisons between two vertices outside that copy are determined by their distances to r . An outside vertex in a higher-scale copy is farther than every vertex in copy i . An outside vertex in a lower-scale copy is at distance $d(u, r) + \epsilon$, with $0 < \epsilon < 100^i$; its position relative to an inside vertex is determined by the sign of the local gap from r . These rules depend only on copy indices and the shared local rankings. They also show that no anchored tie occurs.

It remains to exclude shortcuts through other copies in an arbitrary compatible metric d' . For an outside vertex z and nonroot vertex u of a fixed copy, the ranking input requires $d'(u, z) > d'(u, r)$. If u, v are both nonroot vertices of this copy, then

$$d'(u, z) + d'(z, v) > d'(u, r) + d'(r, v) \geq d'(u, v).$$

If one endpoint is r , the inequality $d'(v, z) > d'(v, r)$ already excludes a tight witness z . Thus no outside vertex is a tight witness for a pair inside the copy.

Every required pair of the restricted five-point metric is therefore required in the full metric. Its five forced edges are forced, and the incompatibility proof forces at least one of its two optional pairs to be required. Each copy contributes at least six distinct edges to any realization, so $s \geq 6k$. The constructed graphs give $s \leq 6k$. Their common edge intersection is precisely the $5k$ local forced edges: each local optional pair is absent under one choice, and no cross-copy pair is an edge of these realizations. This proves the exact number of forced edges as well. $\square$

7.5 The smallest simultaneous-deletion obstruction

Proposition 7.6 (At most four vertices) *Every metric-compatible strict ranking input on at most four vertices has a realization whose edges are precisely its set of forced edges. Consequently, the smallest number of vertices admitting a minimal incompatible deletion set of size at least two is five.*

Proof. For three vertices the globally longest pair can be made tight through the third vertex, and neither remaining pair has a vertex closer to both endpoints, so both are forced. The cases of one or two vertices are immediate.

For four vertices, the proof uses the complete finite certificate in Appendix A. Enumerating the 1,296 ranking inputs and applying Proposition 2.1 leaves 426 metric-compatible ranking inputs in 20 orbits under relabeling. For each representative we give an integer metric whose distance-preserving graph is its proposed forced-edge set. Edges with no vertex closer to both endpoints are forced. For the other proposed forced edges, each possible tight witness is excluded by an explicit nonnegative integer combination of triangle and ranking inequalities. There are only 14 such branches in total. The independent verifier checks the integer identities, the displayed metrics, and completeness of the orbit enumeration. Thus the proposed forced-edge set is the full forced-edge set and is itself realizable in every case.

A minimal incompatible deletion set of size at least two consists entirely of individually optional pairs. Such an incompatible set cannot exist when all pairs outside the forced-edge set can be omitted together. Theorem 7.2 provides the five-vertex example attaining the bound. $\square$

7.6 The largest possible minimum edge count

Theorem 7.7 *For every $n \geq 3$,*

$$\max_{\Pi \text{ metric-compatible}} s(\Pi) = \binom{n}{2} - 1.$$

The maximum is attained by a strict ranking input with $\mathcal{F}(\Pi) = \binom{V}{2} \setminus \{12\}$, while both its HSP graph and RNG are the star centered at 0.

Proof. For the upper bound, take any compatible metric and a globally longest pair uv . This pair is farthest at both endpoints. Increase its distance to

$$t = \min_{z \neq u, v} (d(u, z) + d(z, v)).$$

We have $t \geq d(u, v)$, so the two endpoint orders are preserved. Triangles with uv on the right only become weaker, and those with uv on the left hold by the definition of t . The pair now has a tight witness, giving a realization with at most $\binom{n}{2} - 1$ required edges.

For equality, let $V = \{0, 1, \dots, n-1\}$. At 0 rank the other vertices in increasing numerical order. At each $i > 0$, rank 0 first and all other positive labels in decreasing numerical order. This ranking input is realizable: set $C = 4n$ and define

$$d(0, i) = C + i, \quad d(i, j) = 2C - i - j \quad (1 \leq i < j \leq n-1), \quad d(1, 2) = 2C + 3, \quad (5)$$

where the last assignment overrides the preceding one. All triangle inequalities through 0 hold, with equality for 12. The other leaf distances lie strictly between $2C - 2n$ and $2C$, so every triangle on positive labels also holds. These distances induce the stated strict orders, including at 1 and 2.

Now consider *any* compatible metric, and write $a_i = d(0, i)$. Every edge $0i$ is forced because 0 is nearest to i . A positive label $z \notin \{i, j\}$ cannot be a tight witness for a leaf pair ij :

$$d(i, z) + d(z, j) > a_i + a_j \geq d(i, j).$$

Thus only 0 could witness edge removal. If $i < j$ and $ij \neq 12$, choose $k < j$ with $k \neq i$ and $k > 0$; such a k exists because $j \geq 3$. The orders and triangle inequality give

$$d(i, j) < d(i, k) \leq a_i + a_k < a_i + a_j.$$

Therefore 0 is not a tight witness either. Every pair except 12 is forced. In (5), 12 is removable, so the bound is attained and the asserted forced-edge set is exact.

Finally, at each leaf HSP first selects 0 and screens all other leaves; at 0 it selects every leaf, since no leaf screens another. The RNG has the same star edges, since 0 is closer to both endpoints of every leaf pair. $\square$

This family separates the HSP lower bound from the optimum by $\binom{n}{2} - n = n(n-3)/2$. Theorem 7.5 gives a different separation: even the full set of forced edges may fall short of the optimum.

7.7 Maximizing additive triples can require more edges

For a set S of additive triples, write $D(S) = \{uw : (u, v, w) \in S\}$. Maximizing the number of additive triples means maximizing $|S|$. Minimizing edges means maximizing $|D(S)|$, the number of distinct endpoint pairs with an additive witness. This difference is strict even for complete anchored data.

Theorem 7.8 (More additive triples can require more edges) *There is a strict anchored ranking input on seven vertices for which*

$$\max_{d \models \Pi} |\mathcal{A}(d)| = 19, \quad \min_{\substack{d \models \Pi \\ |\mathcal{A}(d)|=19}} |E(B(d))| = 10,$$

but

$$s(\Pi) = 9, \quad \max_{\substack{d \models \Pi \\ |E(B(d))|=9}} |\mathcal{A}(d)| = 18.$$

Thus the two objective functions have disjoint sets of optimizers.

Certificate proof. The ranking input is

$$\begin{array}{c|cc|c} 0 & (5, 3, 1, 2, 4, 6) & 1 & (4, 6, 2, 5, 0, 3) \\ 2 & (1, 5, 4, 6, 0, 3) & 3 & (6, 0, 4, 1, 5, 2) \\ 4 & (6, 1, 2, 5, 3, 0) & 5 & (0, 2, 4, 1, 6, 3) \\ 6 & (4, 1, 3, 2, 5, 0) & & \end{array}$$

Two integer metric witnesses are given in Table 1. The nine-edge distance-preserving graph is

$$G_9 = \{01, 03, 05, 12, 14, 25, 36, 45, 46\},$$

and the ten-edge graph is $G_{10} = G_9 \cup \{15\}$. Direct evaluation gives 18 and 19 additive triples, respectively, establishing attainment.

For the upper bounds, the supplementary file `hybrid_objective_separation.json` gives exact certificates for the lexicographic maxima

$$\max(|\mathcal{A}|, |D(\mathcal{A})|) = (19, 11), \quad \max(|D(\mathcal{A})|, |\mathcal{A}|) = (12, 18).$$

Appendix B specifies the complete verification procedure. Its 34 rational Farkas identities and a proof with 55 search nodes exclude every additive pattern improving either objective. The independent standard-library verifier checks the identities, the realizing metrics, and all search branches without calling a numerical solver. Since $\binom{7}{2} = 21$, these two certified bounds are exactly the claims of the theorem. $\square$

Pair	01	02	03	04	05	06	12
Triples first	30	32	22	37	14	42	14
Edges first	28	29	20	34	14	39	12
Pair	13	14	15	16	23	24	25
Triples first	32	7	26	12	46	21	18
Edges first	30	6	27	11	42	18	15
Pair	26	34	35	36	45	46	56
Triples first	26	25	36	20	23	5	28
Edges first	23	24	34	19	23	5	28

Table 1: Integer witnesses for Theorem 7.8. The first is twice the rational witness stored in the certificate file.

The two witnesses share 17 additive triples. The triples-first witness additionally has 054, 056, whose endpoint pairs already have other tight witnesses. The edge-first witness instead has 125, permitting removal of 15. Deleting 15 from the displayed ten-edge weighting without reweighting raises d_{15} from 26 to 30 and ties $d_{10} = 30$, violating the ranking at 1. The improvement in edge count therefore requires a different compatible metric. No minimality claim is made for seven objects, and the theorem’s upper bounds are computer-assisted.

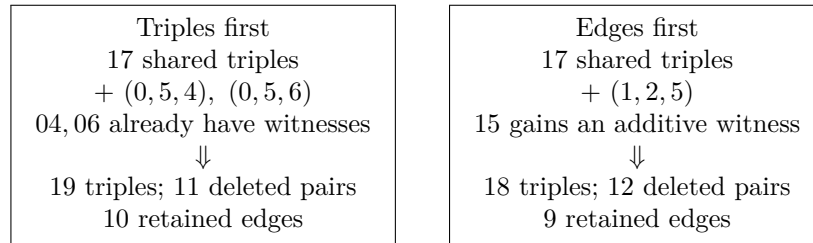


Figure 3: The witnesses in Table 1. The shared triples already include (0, 1, 4) and (0, 1, 6), accounting for endpoint pairs 04 and 06. Two additional witnesses for those pairs increase the triple count without removing another edge. One witness for the new pair 15 instead reduces the edge count. The metrics differ; this is not deletion at fixed weights.

7.8 Other density and complexity observations

Proposition 7.9 (An explicit dense family) *For every $n \geq 4$ there is a strict ranking input with $s(\Pi) = \lfloor n^2/4 \rfloor$.*

Proof. Let A, B be a balanced partition of V and enumerate the $m = |A||B|$ edges of $K_{A,B}$ as $e_0, \dots, e_{m-1}$. Give e_j length $1 + 2^j/2^{m+1}$. All these lengths lie in $(1, 5/4]$. Every cross-part shortest path is its direct edge, and every same-part shortest path uses two edges: longer alternatives use at least three and four edges, respectively. Binary uniqueness of the perturbation sums ensures strict anchored rankings, because distinct simple paths from a fixed source to different endpoints have different edge sets.

All cross distances are less than 2 and all same-part distances exceed 2. Thus no cross pair has a vertex closer to both endpoints, while every same-part pair has such a vertex on a shortest two-edge path. The RNG is exactly $K_{A,B}$. Every RNG edge is required in every compatible metric, since a tight witness would be closer to both endpoints. This forces all m cross edges; the construction attains m . $\square$

The upper-bound operation in Theorem 7.7 applies simultaneously to all reciprocal-farthest pairs: strict rankings make them a matching, and each triangle contains at most one modified pair. Together with Proposition 2.1, this constructs a compatible graph with those pairs removed in $O(n^2)$ time. The bipartite family above has a different feature from the extremal family: its RNG already forces the optimum.

Proposition 7.10 (Decision membership in NP) *The problems of deciding whether $s(\Pi) \leq b$ or whether a compatible metric has at least a additive triples belong to NP.*

Proof. Guess a set S of at least $\binom{n}{2} - b$ omitted pairs and a tight witness z for each. Require $d(u, v) \geq 1$ for every distinct pair, unit lower bounds on consecutive ordinal gaps, all triangle inequalities, and the guessed tight equalities. There are finitely many positive distances and strict gaps, so one scaling normalizes both collections simultaneously, including when the collection of rank gaps is empty. The resulting feasible set is a closed pointed rational polyhedron in $\binom{n}{2}$ variables, with polynomially many rows and bounded integer coefficients. If nonempty, minimizing the sum of the pair distances has a compact nonempty sublevel set and attains a vertex. A determinant bound therefore gives a rational solution of polynomial encoding length. Such a solution and the guessed witnesses can be checked in polynomial time; Proposition 7.1 proves correctness. For the additivity threshold, guess a triples and use the same normalized linear feasibility system. $\square$

NP-hardness, a polynomial algorithm, and useful approximation guarantees for this general optimization problem remain unresolved here.

8 Computational checks and running times

The computational claims have two parts: a witness establishes a feasible realization, while an obstruction or complete branch-by-branch proof excludes the alternatives needed for rejection or optimality. The measurements then describe the cost of producing and checking such evidence in the specified implementations.

The accompanying `ordinal_tree_research.py` separates numerical search from exact verification. It uses SciPy [30] with HiGHS [29] to find a candidate primal or dual; acceptance is recorded only after converting to rational or integer values and checking all inequalities exactly. A weight rejection requires positive integer multipliers whose weighted row sum is exactly zero. Failure to verify a numerical result raises an error. This reference solver is not an implementation of Tardos's algorithm and makes no strongly polynomial performance claim.

Exact rational LP software addresses the numerical step more systematically. Eifler, Nicolas-Thouvenin, and Gleixner [28] combine LP iterative refinement with precision boosting, using increased precision when approximate solutions at the current precision are inadequate. Our proto-

type instead reconstructs candidates and checks their rational identities; it may report unresolved reconstruction. A valid returned certificate proves the corresponding claim, but failure to reconstruct one is not an infeasibility result. The exact-LP oracle used in the complexity statements and Appendix B is therefore distinct from this practical search-and-check implementation.

We enumerated all $(3!)^4 = 1,296$ complete strict ranking inputs on four labels. The classification is

Outcome	Inputs
No compatible metric (pair-order cycle)	870
Metric-compatible, HSP rejects candidate edges	186
Metric-compatible, candidate edges pass, weights fail	120
Positive tree realization	120

For each of the 426 metric-compatible ranking inputs, all 16 labeled trees were also tested using an independent edge-path-incidence formulation. All 6,816 of these LP decisions received exact witnesses, and the acceptance and uniqueness results agreed with HSP followed by the root-distance LP. The 120 weight rejections had extracted supports of size two (89 cases) or three (31 cases); these counts describe the extracted certificates and do not claim minimum support.

For every labeled tree through six vertices, every root and every ordered triple were used to audit (1). All 972,210 rows had support at most four, coefficient magnitude at most two, and squared norm at most ten. These tests supplement the proof of Lemma 4.1.

Four tree families were checked: paths, stars, balanced binary trees, and seeded Prüfer trees. The sizes were 4, 8, 16, 32, 64, and 128. Distinct powers of two give exact strict ranking inputs. All edge sets, inequalities in root-distance variables and nonedge certificates were verified. At $n = 128$:

Family	Comparison queries	Matrix nonzeros	Recorded comparisons
Path	47,982	32,903	16,002
Star	48,006	32,256	16,002
Balanced binary	47,898	35,396	16,002
Prüfer	48,974	35,197	16,002

The last column counts the partial observations used to exclude all nonedges, as in Corollary 6.3. Each matrix had 16,255 rows, and the query ceiling is $4 \cdot 127 \cdot 126 = 64,008$. Separately, 40 randomly weighted trees through $n = 32$ were reconstructed using LP-discovered weights, with exact primal verification. The two five-vertex metrics, all 2^k choices for the composition at $1 \leq k \leq 5$, and the dense construction for $4 \leq n \leq 20$ were also checked using exact integer distances. The environment, seeds, individual counts, and observed runtimes are in `audit_results.json`. The extremal family in Theorem 7.7 was checked at all sizes $3 \leq n \leq 32$ and at $n = 64, 128$, including every ordinal witness used in the forcing proof. These finite checks supplement its algebraic proof. The separate four-point deletion classification gives 120 ranking inputs of optimum three, 234 of optimum four, and 72 of optimum five. Its 20 integer metric witnesses and 14 Farkas certificates are checked by a standalone program using only the Python standard library; no numerical solver is used in that verification.

These are correctness and reproducibility checks. They do not establish asymptotic solver performance, statistical validity, or a general HSP speed advantage. No empirical psychophysical dataset was fitted in this study.

8.1 Tree recovery, weight testing, and additive-triple counts

The separate integer verifier `verify_hsp_additivity.py` checks 1,441 labeled trees through six vertices, their 81,711 endpoint–middle triples, and 58,578 exclusions of vertices outside the endpoint path. It also checks the two five-vertex additivity examples, including their exclusions and exact maximizing patterns.

The earlier studies timed only edge recovery. Their records are retained in `hybrid_graph_results.json` (five-repeat HSP medians) and `tree_strategy_results.json` (three-repeat HSP versus anchored forbidden-triple comparison, 16 prespecified ranking inputs). The latter implementations both return the tree edges and the maximum number of additive triples; the forbidden-triple implementation explicitly enumerates exclusions in $O(n^4)$. Its slower timings do not establish an advantage over the quadratic metric–MST baseline. The end-to-end comparison below supplies that missing control and includes the weight stage.

End-to-end comparison with metric–MST. The prespecified positive suite has four families (path, star, balanced binary, seeded Prüfer) at $n = 16, 32, 64, 128$, with seed 20260911. Integer edge lengths have random integer parts in $1, \dots, 100$ and distinct binary residues to forbid ties. Only the resulting rank lists enter either recognizer. Both pipelines include the same pair-order consistency test and compatible metric construction. The HSP pipeline then uses HSP screening with a selection limit; the MST pipeline uses dense Prim. They use identical root-distance row construction and HiGHS feasibility code. Algorithm 1 does not require a synthetic metric: this standardized comparison measures the stated pipelines, not the minimum preprocessing cost achievable by either method. On acceptance, rational extraction, integral scaling, checking every inequality in root-distance variables, and independent recomputation of tree distances and rankings are all timed. Imports and ranking input generation are excluded.

Each method receives one warm-up and three timed repetitions per case, alternating execution order. The primary LP has a 30-second time limit; all 96 timed positive runs completed and reconstructed the generating edge set. Table 2 reports the largest size; all stages, individual repetitions, inputs, and environment versions are stored in `tree_recognition_results.json`. Stage medians need not sum to the median total. The total column also gives the observed minimum and maximum of the three runs, not a confidence interval; all four pairs of total-time ranges overlap at $n = 128$. No consistent end-to-end winner emerges. The shared matrix, LP, and exact verification stages dominate these accepted instances, so timing only edge recovery would give an incomplete practical comparison.

The independent metric–MST check recovers all 1,441 labeled trees through six vertices using each of three compatible-metric constructions (4,323 checks). Exhaustive four-vertex recognition gives the same 120 accepted ranking inputs for both pipelines. Of the remaining ranking inputs, 870 fail the common consistency test. HSP rejects 186 at edge recovery and 120 at weight feasibility; MST rejects all 306 at weight feasibility. This demonstrates an early-rejection benefit, not an accepted-instance speed guarantee. Twelve rejection controls, the first four ranking inputs in each HSP rejection class in lexicographic enumeration, also receive three timed runs per method. All 72 complete with the expected decision. For weight rejection, the recorded exact stage includes auxiliary primal/dual certificate search as well as rational checking; it is not a pure verifier timing. There are no unresolved timed runs in this suite.

The supplement stores one standalone decision certificate per case in `benchmark_decision_certificates.json`: 16 integer tree weightings, eight Gordan certificates on the metric–MST candidate, and four explicit pair-order cycles. The positive weights are regenerated from the published instance seed, not recovered outputs of the historical timed solver runs; the negative certificates

Family	Method	Metric	Edges	Matrix	LP	Exact	Total
Path	HSP	23.5	6.7	139.5	155.9	211.7	537.9 [508.2, 584.6]
Path	MST	27.0	3.7	135.7	149.7	174.1	499.6 [441.0, 516.2]
Star	HSP	23.1	7.9	132.1	56.2	174.9	392.7 [319.4, 416.3]
Star	MST	34.2	4.9	143.2	58.8	156.6	374.9 [362.6, 472.6]
Binary	HSP	31.0	6.4	131.4	54.9	204.0	429.2 [350.9, 442.2]
Binary	MST	19.5	2.9	108.3	51.7	186.2	358.6 [330.1, 478.4]
Prüfer	HSP	24.7	7.8	147.4	71.9	143.1	419.3 [289.0, 510.6]
Prüfer	MST	32.0	3.8	143.5	84.5	236.6	500.4 [263.1, 558.1]

Table 2: Complete recognition at $n = 128$: median milliseconds over three runs. Metric includes the common consistency check and synthetic metric construction; Edges is the time spent finding candidate tree edges. Matrix includes LCA preprocessing and sparse solver conversion. Exact includes primal extraction and validation against the original rankings. Total entries are median [minimum, maximum].

are computed separately with exact arithmetic. `verify_benchmark_certificates.py` replays all 28 decisions without numerical optimization and checks every recorded timing median. Full stage-by-stage observed ranges are in `tree_timing_variability.json`.

8.2 Comparing graph optimization methods

The reference searches in Appendix B compare two objectives on the same anchored rankings: maximize the number of additive triples; or minimize edges, breaking ties by the larger number of additive triples. In the separate seven-vertex example, we also break ties for the first objective by the smaller edge count. The direct baseline uses include/exclude branching, joint LP feasibility, and a monotone objective bound. The comparison implementation adds HSP filtering, a forward greedy feasible family, and learned-conflict branching. Both reconstruct rational primal witnesses and Farkas certificates and check them exactly. Neither implementation is Klauer’s original program or a tuned modern mixed integer solver.

The paired suite contains 11 ranking inputs: the two five-vertex examples, four generated five-vertex ranking inputs, four generated six-vertex ranking inputs, and one seven-vertex ranking input. Both objectives are optimized on every ranking input, yielding 22 paired comparisons with identical optimum scores. Generated instances start from a seeded random labeled spanning tree, add edges independently with probability $0.15 + 0.2j$, $j = 0, 1, 2, 3$, and assign positive integer weights with distinct binary perturbations. Only shortest-path rankings enter the optimizer. The seed is 20260909. The seven-vertex paired case uses $j = 0$. A pilot direct run on another seven-vertex case became slow and was interrupted; the paired suite was limited to the completed case. Thus the timing sample is exploratory and completion-selected, not a prespecified representative scaling study.

The separate comparison of search components uses the original five-vertex ranking input and three six-vertex ranking inputs (seed 902026, added-edge probability 0.3), with the edge-first objective. Table 3 shows that the benefit here comes from conflict-directed branching: HSP excludes no additional candidates on these four instances, and the forward greedy pass increases the number of LPs. Other paired instances do benefit from HSP filtering. On one easy five-vertex paired case the direct method uses six feasibility LPs while the combined version uses eight. A conflict certifies a whole incompatible set and lets the search prune many candidate patterns. Greedy preprocessing instead pays for extra LPs to improve the best feasible solution found so far; if the upper-bound

proof is already small, that cost need not be recovered. These observations explain the component comparison but do not support a uniform benefit from greedy preprocessing.

Instance	Direct alone	Direct with HSP	Conflict search with HSP	Greedy and conflicts, HSP
Five vertices	98	98	7	11
Six vertices, input 0	190	190	7	20
Six vertices, input 1	22	22	6	20
Six vertices, input 2	195	195	9	20

Table 3: Feasibility LP calls when minimizing edges, with ties broken by the larger number of additive triples. Each column uses the stated search components.

An additional seeded search compared both lexicographic objectives on 63 generated ranking inputs before finding Theorem 7.8; it is a counterexample search, not an estimate of the frequency of separation. The saved proof audit covers 68 records: 67 claimed optima and one deliberately suboptimal greedy example. The independent checker verifies 5,424 rational Farkas certificates and 540 certified search nodes. The seven-vertex theorem alone uses 34 certificates and 55 search nodes. The robustness regression additionally rejects six classes of altered proof input under normal, optimized, and doubly optimized Python execution (18 rejection checks), while accepting the original in all three modes.

Python 3.12, SciPy 1.18.1, and SymPy 1.14.0 were used for the new graph searches; the JSON files record the exact runtime versions. Timings are single-run exploratory measurements for the graph studies. The older tree studies use five and three repeats, respectively; the new complete recognition study uses three repeats. No graph timing claim is promoted to a performance contribution. Numerical witnesses and times can vary by machine or solver; verified scores and certificate validity are the reproducibility targets. There are no statistical generalization, approximation-ratio, or broad solver-performance claims from this suite.

9 Scope and next questions

For trees, the results separate edge recovery from weight feasibility. Complete anchored rankings determine at most one possible edge set, and a linear program tests its weights. When weights exist, all individually possible additive triples hold in a realizing tree metric. For general graphs, individually possible edge deletions need not be possible together.

One question is whether the root-distance inequalities, with at most four nonzero coefficients per row, admit a substantially faster specialized algorithm. Another is to close the gap between the $\Omega(n \log n)$ lower bound and $O(n^2)$ upper bound on comparison queries for recovering a promised tree. For partial comparisons, Theorem 6.2 gives a sufficient condition for recognition, and Corollary 6.4 bounds a search over candidate trees. Neither settles recognition from arbitrary partial comparisons.

For graphs, the complexity of minimizing edges remains unresolved here. Useful questions include the possible gap between the two objectives, the size and structure of minimal incompatible deletion sets, and parameters that reduce the search over additive equalities. Theorem 7.7 determines the largest possible minimum edge count. Proposition 7.6 determines the smallest number of vertices for which individually possible deletions can be incompatible; a short structural proof in place of the finite classification would be useful. Theorem 7.8 shows that the two objectives can have different optimizers, but a smaller example or a shorter structural proof remains open

here. For cycles and cactus graphs, one can ask which anchored rankings admit positive edge weights. Numerical cactus realization and circular seriation [18, 19] provide related results, but do not answer this anchored-comparison question. Smaller proofs of graph optimality are another algorithmic target.

Finally, exact realization is different from statistical model selection. The local tests of Victor et al. [23] motivate collecting anchored comparisons. A rejection certificate here rules out a positive tree whose vertices are exactly the observed objects; it does not rule out trees with unobserved branch vertices. A numerical solver’s failure to produce an exactly checkable answer is not a rejection certificate.

A Finite certificates for four vertices

Table 4 lists all metric-compatible four-vertex ranking inputs up to relabeling. A string such as 023 denotes an increasing distance order. The row at anchor 0 is 123 throughout. The distance vector uses the order (01, 02, 03, 12, 13, 23), and m is the orbit size. Each displayed integer metric has distance-preserving graph exactly C .

ID	π_1	π_2	π_3	Distance vector	C	m
1	023	013	012	(1, 2, 3, 3, 4, 5)	{01, 02, 03}	24
2	023	013	021	(2, 3, 5, 5, 7, 6)	{01, 02, 03, 23}	24
3	023	013	102	(1, 2, 5, 3, 4, 7)	{01, 02, 13}	24
4	023	013	120	(2, 3, 8, 5, 6, 7)	{01, 02, 13, 23}	24
5	023	013	201	(1, 2, 6, 3, 7, 4)	{01, 02, 23}	24
6	023	013	210	(1, 2, 6, 3, 5, 4)	{01, 02, 13, 23}	24
7	023	031	021	(3, 4, 5, 7, 8, 6)	{01, 02, 03, 23}	24
8	023	031	201	(2, 3, 7, 5, 9, 4)	{01, 02, 23}	24
9	023	031	210	(2, 4, 9, 6, 7, 5)	{01, 02, 13, 23}	24
10	023	301	201	(1, 2, 3, 3, 4, 1)	{01, 02, 23}	12
11	023	301	210	(1, 4, 7, 5, 6, 3)	{01, 02, 13, 23}	24
12	032	013	012	(3, 4, 5, 7, 6, 9)	{01, 02, 03, 13}	24
13	032	013	102	(1, 2, 3, 3, 2, 5)	{01, 02, 13}	12
14	032	013	120	(3, 4, 9, 7, 6, 8)	{01, 02, 13, 23}	24
15	032	031	012	(4, 5, 6, 9, 7, 8)	{01, 02, 03, 13, 23}	24
16	032	031	021	(4, 5, 6, 9, 8, 7)	{01, 02, 03, 13, 23}	24
17	032	031	120	(2, 3, 5, 5, 3, 4)	{01, 02, 13, 23}	12
18	032	031	201	(4, 5, 7, 9, 8, 6)	{01, 02, 03, 13, 23}	24
19	032	031	210	(3, 4, 9, 7, 6, 5)	{01, 02, 13, 23}	24
20	032	301	210	(1, 2, 3, 3, 2, 1)	{01, 02, 13, 23}	6

Table 4. The 20 representatives and integer witnesses.

For completeness of the lower bounds, write all inequalities as $R_j d \geq b_j$. Rows 0–5 are distance positivity in lexicographic pair order; rows 6–13 are consecutive rank differences, ordered by anchor and position; rows 14–25 are triangle slacks $d(u, z) + d(z, v) - d(u, v) \geq 0$, ordered by pair uv and then z . Thus $b_j = 1$ for $j < 14$ and $b_j = 0$ otherwise. This normalization is equivalent to strict positivity and strict ranking gaps by scaling.

A tight witness z for a proposed removable pair uv must be closer to both endpoints than they are to each other. If no such z appears in the ranking input, the edge is forced immediately. Otherwise append the negative of its triangle row as row 26 with right-hand side zero. A certificate

gives nonnegative integers λ_j such that

$$\sum_j \lambda_j R_j = 0, \quad \sum_j \lambda_j b_j > 0.$$

Summing the inequalities is a contradiction. Across all proposed sets of forced edges, only 14 witness branches need this treatment, and every stored multiplier is one. For example, in row 2 the only possible witness for 23 is 0. If it were tight, then

$$d(0, 2) + d(0, 3) = d(2, 3) < d(1, 3) \leq d(0, 1) + d(0, 3),$$

contrary to $d(0, 1) < d(0, 2)$.

The file `four_point_certificates.json` records every branch and its constraint indices. The independent program `verify_four_point_certificates.py` checks all integer identities, all witness metrics, all possible witness branches, and the exhaustive enumeration and orbit sizes. It imports no optimization or scientific package. Certificate generation is separate, in `generate_four_point_certificates.py`. In particular, the finite proof does not treat a floating-point infeasibility status as evidence.

B Exact graph search and proofs of optimality

A realizing metric proves attainment of an objective value, but not its optimality. This appendix specifies the reference method that completes the graph certificates by covering every improving additive pattern. It specializes the established maximum feasible subsystem and conflict-cover approach [24]; its purpose here is a precise, independently checkable computation. Begin with the acyclicity test of Proposition 2.1 and reject metric-inconsistent input. Throughout the search below, Π has passed this test, so the base system with no selected additive equalities is feasible.

Let Q contain all triples (u, v, w) , with canonical endpoint order, satisfying $v \prec_u w$ and $v \prec_w u$, except those whose endpoint pair is an HSP edge. By positivity and Theorem 5.2, $\mathcal{A}(d) \subseteq Q$ for every compatible metric. Write $q = |Q|$. Every unordered three-set has at most one such orientation, so $q \leq \binom{n}{3}$. Further sound forbidden-triple tests may be applied, but are not necessary for correctness.

For $S \subseteq Q$, joint feasibility is the following linear system:

$$\begin{aligned} d_{uw} &\geq 1 & (u < w), \\ d_{o, \pi_o(i+1)} - d_{o, \pi_o(i)} &\geq 1 & (\text{successive ranks}), \\ d_{uv} + d_{vw} - d_{uw} &\geq 0 & (\text{all distinct triples}), \\ d_{uv} + d_{vw} - d_{uw} &= 0 & ((u, v, w) \in S). \end{aligned} \tag{6}$$

Any compatible finite metric can be scaled so that both its positive distances and its strict rank gaps are at least one. Hence the normalization loses no feasible equality pattern. No finite upper distance bound or uncertified big-M constant is used.

Represent each selected equality by its two opposite weak inequalities and write the resulting system as $Cx \geq b$. An infeasibility certificate is

$$\lambda \geq 0, \quad C^T \lambda = 0, \quad b^T \lambda = 1.$$

The selected equality indices with positive multiplier support form a conflict $K \subseteq S$. The fixed base system is feasible because the input passed the metric-consistency test, so K is nonempty. Every

feasible equality family omits at least one member of K . We call this set a *conflict*: its equalities cannot all hold in one compatible metric.

To maximize additive triples use $f(S) = |S|$; to minimize edges use $f(S) = |D(S)|$. The reference method also supports either lexicographic ordering of these two monotone scores. At a node, let $R \subseteq Q$ be the allowed family and let L be the score of the best verified metric found so far (the *incumbent*). If $f(R) \leq L$, prune. Otherwise test (6) for $S = R$. If feasible, its metric attains at least $f(R)$ and updates the incumbent. If infeasible with conflict K , recurse on $R \setminus \{t\}$ for each $t \in K$. Cache visited families and conflicts. A forward greedy feasibility pass is optional and supplies only an incumbent.

Proposition B.1 (Exact search) *For a metric-compatible complete strict ranking input Π , with an exact LP subroutine the search returns an optimal realization for either score or either lexicographic combination in $2^q \text{poly}(n)$ time.*

Proof. Every compatible set of additive triples is a subset of Q . At an infeasible node it belongs to at least one child because it omits a conflict member. Monotonicity makes $f(R)$ an upper bound for patterns contained in R . A feasible node attains that bound or better as a global incumbent. There are at most 2^q distinct families and polynomial work per family, including $O(n^3)$ -size rational LPs and conflict extraction. A witness can satisfy extra equalities outside R ; using it as a global incumbent is safe and does not assert that its full pattern belongs to that branch. $\square$

The bound is fixed-parameter tractability in q , not polynomial time in n for unrestricted graphs. The prototype calls HiGHS for a numerical candidate, reconstructs it rationally, and checks all identities. It raises an error when reconstruction is unresolved; it is not a worst-case exact LP implementation. The refinement and precision-boosting approach of [28] concerns obtaining exact LP answers; the branch-by-branch proof below concerns how such answers certify a discrete optimum. Our checker uses the resulting rational identities independently of the search solver.

Independent verification. The JSON records contain the ranking input, candidate ordering, rational metric, claimed score, and, for each conflict, the selected equalities, supported row indices, rational multipliers, and equality indices involved. The checker reconstructs every row of (6) and verifies all three Farkas conditions exactly. It verifies that every excluded ordinally eligible candidate has an HSP endpoint pair. It then starts at Q , prunes a family only when its score is no better than the claimed optimum, and otherwise splits on a contained certified conflict. If no such conflict exists, verification fails. This constructs a proof covering all improving patterns without relying on the search program’s status. The actual metric provides the matching lower bound.

The compact check of Theorem 7.8 is

```
python verify_hybrid_results.py hybrid_objective_separation.json
```

All decisive verifier checks use explicit exceptions and remain active under `python -O` and `python -OO`. The accompanying `test_verifier_robustness.py` accepts the original proof and rejects six altered inputs in each of the three execution modes: changed multiplier, invalid support, missing cover, changed score, invalid metric, and duplicated candidate. The compact theorem checker `verify_hybrid_results.py`, with the displayed input file, verifies two objective records, 34 Farkas identities, and 55 search nodes. Running the verifier without an argument checks the full saved suite. Both modes require only the Python standard library and the accompanying ordinal helper module. Proof size and verifier time can be exponential; a short certificate of global optimality for every instance is not asserted.

C Complete ranking inputs with ties

Complete rankings with ties specify ordered tie blocks at each anchor. In HSP, choose an arbitrary active minimum, but screen x only on the *strict* comparison $d(x, a) < d(x, o)$; do not screen merely because a tie was broken in the traversal order.

Proposition C.1 *Whether complete rankings with ties admit a positive tree on the observed vertices can be decided in strongly polynomial time. The candidate tree is unique. Given that edge set, an infeasible system admits a certificate using at most n strict-comparison/positivity and equality rows in total.*

Proof. The branch proof of Theorem 3.1 does not require distinct distances in different branches: the adjacent vertex is still the unique minimum in its own branch, screening within a branch is strict, and no cross-branch screening occurs. The same global selection limit therefore identifies the tree with $O(n^2)$ work.

Within each ordered tie block, connect its elements by consecutive equalities; between adjacent blocks impose one strict comparison between representatives. Along with positivity, these give $Ah > 0$, $Eh = 0$ with $O(n^2)$ bounded rows with at most four nonzero coefficients. Normalize the strict rows to $Ah \geq 1$; standard-form bounded-matrix LP proves the complexity claim.

Let $p = n - 1$ and $t = \text{rank } E$. Restrict the strict rows to $\ker E$, which has dimension $p - t$. If infeasible, Gordan’s alternative and Carathéodory give a positive dependence on at most $p - t + 1$ projected strict rows. Their weighted sum in the original coordinates belongs to the row space of E , so it can be canceled by at most t independent equality rows with unrestricted multipliers. This uses at most $p + 1 = n$ rows altogether. Rationality follows by applying the alternative to a rational basis for the nullspace. $\square$

This appendix concerns exact ties and positive edges. Zero-length edges would destroy edge set uniqueness. The supplied reference recognizer implements the case of complete strict rankings only.

D Repeatable independent noise

Assume each repeated raw answer is independently correct with probability at least $1/2 + \alpha$, conditional on the requested comparison and all past queries. Use an odd number r of independent repetitions and majority vote for each logical query. Hoeffding’s inequality gives error probability at most $\exp(-2\alpha^2 r)$. Run the oracle HSP procedure with the global selection limit. Even erroneous answer histories make fewer than $4n^2$ queries: at most $2n - 1$ tournaments and $2n - 2$ screenings occur, each with at most $n - 2$ comparisons. Thus

$$r \geq \frac{1}{2\alpha^2} \log \frac{4n^2}{\delta}$$

(rounded upward to an odd integer) makes every logical answer correct with probability at least $1 - \delta$, by conditional error bounds and a union bound. Under the tree promise this recovers the true edge set using $O(n^2 \alpha^{-2} \log(n/\delta))$ raw judgments. The argument is a standard noise-amplification reduction; it does not address persistent bias, dependent repeats, or calibrated goodness of fit.

Code and data availability

The source package accompanying this manuscript includes the programs, exact certificates, and recorded synthetic data as ancillary material. The original tree audit uses `ordinal_tree_`

research.py. The independent programs `verify_four_point_certificates.py`, `verify_hsp_additivity.py`, and `verify_hybrid_results.py` check the four-vertex classification, additive triples, and graph optima, respectively. The programs `hybrid_graph_optimization.py` and `hybrid_followup_checks.py` run graph searches and compare their components. The ancillary README gives dependencies and commands; the files `manifest.json` and `verify_manifest.py` support integrity checks. The computations and recorded measurements are those of proof release 2026.09.13-r1, including the metric-MST baseline, complete recognition timings, and optimized-execution checks. This revision updates the exposition and program documentation. The ancillary EXPERIMENTS.md presents the recorded results using the terms in this paper. Proposition 7.6 and Theorem 7.8 rely on the supplied finite certificates, which should be preserved with the article and its source.

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